

Gadi Thanmayee

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[Portfolio](#)

Objective :

Curious and motivated 3rd-year engineering student with a strong focus on Machine Learning, Generative AI, Computer Vision, and IoT microcontrollers. Passionate about learning new technologies and applying them to real-world projects. Eager to gain hands-on experience through internships, contribute to meaningful projects, and develop practical skills while exploring innovative solutions.

Education :

- Marri Laxman Reddy Institute of Technology and Management [MLRITM] , Btech - Sept 2024 – Jun 2028
Computer Science and Engineering [AIML]
- Intermediate , Narayana Junior College , Kukatpally July 2024

Projects :

Plant Health AI Platform : Developed a multimodal leaf disease diagnostic platform using transfer learning (CNNs) to classify crop pathologies, providing calibrated confidence scores, disease treatment guidance, and a web dashboard for agricultural decision support.

Ultron Autonomous Desktop Assistant : Developed an autonomous desktop agent translating voice and natural-language commands into precise operating system actions, application navigation, active system telemetry (CPU/RAM), and workflow automation using Python and PyAutoGUI.

IntelliDoc RAG Engine : Built an enterprise Retrieval-Augmented Generation application integrating MongoDB Vector Search and Sentence Transformers to index technical manuals, hardware datasheets, and answer complex domain queries with zero hallucinations.

IoT-Based Smart Agricultural System : Built a basic smart agriculture system using sensors to monitor environmental conditions and support simple irrigation decisions using IoT concepts.

Spam Email Detection Using Machine Learning : Created a basic supervised machine learning model to classify emails as spam or non-spam, involving data preprocessing, feature extraction, and model evaluation using customized 13 features and connected to an email for real time detection.

IoT Chatbot : Built a RAG-based chatbot for IoT sensors and microcontrollers that retrieves relevant information from technical documents, integrates Wikipedia API for additional knowledge, generates context-aware responses using an LLM, and showcases results through a Flask Interface.

Accident Detection using Deep Learning : Developed an AI-based system to detect road accidents from images using deep learning. Built and trained the model using Python, TensorFlow/Keras, NumPy, and Pandas, and deployed it through a Flask-based web application for accident detection.

Deepfake Detection using Deep Learning : Developed a deep learning model to identify AI-generated or manipulated media (deepfakes). Implemented the detection system using Python, TensorFlow/PyTorch, and OpenCV to classify media as real or fake.

YTBT Cardio NLP : Developed YTBT Cardio NLP, a PyTorch 2D CNN + Transformer system translating 12-lead ECG waveform images into structured reports. Built a multi-role Flask portal (Doctor/Patient/Technician/Admin) with SQLite audit logs.

StudyMind AI : Developed StudyMind AI, an adaptive study navigator that generates personalized learning roadmaps and dynamic study timetables using Flask, SQLite, and Google Gemini API. Implemented a Random Forest ML engine to classify student concept mastery and automatically inject remedial topics, secured by a custom proctored, anti-cheat exam portal and synced peer timers.

Skills

- Programming: Python, C (basics), SQL, JavaScript (Basics)
- Core Concepts: Basics of Machine Learning, Supervised Learning, Data Preprocessing
- IoT & Embedded (Basic): Arduino, ESP8266, Sensors, IoT fundamentals
- Libraries & Tools: Pandas, NumPy, Scikit-learn
- Generative AI & RAG: RAG Applications, Prompt Engineering, LLM Integration, Vector Search
- Deep Learning & Frameworks: TensorFlow/Keras, PyTorch, OpenCV
- Backend & Cloud: Flask, SQLite, MongoDB Atlas Vector Search, AWS Cloud Foundations
- Other : Enthusiastic, Problem-solving, willingness to learn, basic project implementation, technical leadership

Experience :

AI/ML Intern

[QuGates Technologies](#) | Bengaluru, Karnataka

June 2026 – Present

- Developing AI and Machine Learning solutions for crop disease detection using computer vision and deep learning techniques.
- Building intelligent image classification models to identify plant diseases and improve agricultural decision-making.
- Working on software automation solutions to streamline workflows and reduce manual processes.
- Performing data preprocessing, model training, testing, and performance evaluation for AI applications.
- Collaborating with the development team to integrate AI models into software solutions.

Part-Time:

[Oorjith](#) | Chandigarh , Punjab

July 2026 – Present

- Develop and support machine learning solutions by applying data preprocessing, model development, and performance evaluation techniques.
- Collaborate with cross-functional teams to build and optimize AI-driven applications using Python and machine learning frameworks.
- Assist in analyzing datasets, improving model performance, and implementing practical AI solutions for real-world use cases.

Patent :

Team Lead — Indian Patent Application No: 202641109078 (IP India)

Title: Uncertainty-Aware Multimodal Crop Health Detection, Forecasting and Adaptive Management System

- Spearheaded an engineering team to architect an uncertainty-aware vision pipeline fusing multispectral crop imagery with environmental sensor metrics to detect crop pathology and output calibrated risk forecasting.
- Formulated an adaptive management loop that maps diagnostic outputs to localized remediation strategies with quantifiable confidence intervals.

Extra-Curricular Activities & Leadership :

- Secretary — Robotics & Automation Lab (RAL Club), MLRITM (2024 – Present): Directing club administrative operations, managing budgets, and coordinating inter-departmental innovation projects across robotics, AI, and embedded automation.
- Lead Event Coordinator — RAL Club Events & Technical Symposia: Organized and orchestrated flagship college robotics exhibitions, technical hack days, and project showcases engaging 200+ active student participants across departments.
- Technical Mentor & Workshop Instructor: Conducted hands-on technical workshops instructing junior batches in Arduino microcontrollers, ESP8266 IoT connectivity, and Python for AI, guiding 100+ students through circuit design and sensor telemetry.
- Hackathon Coordinator & Team Lead — Smart India Hackathon (SIH) & Competitions: Led cross-functional teams in competitive national hackathons (including SIH initiatives) focused on AI-driven agricultural solutions and smart automation prototypes, driving architecture design and live demonstrations.

Courses & Certifications:

- Prime 2.0: AI/ML Batch — Currently Pursuing
Topics: Machine Learning, Deep Learning, Generative AI, RAG, AI Projects, PyTorch, and AI Engineering.
- Python with AI Course — AI for Techies — Currently Pursuing
Focused on Python programming, AI fundamentals, problem-solving, and AI-based application development.
- AI Projects Module — Currently Pursuing
Hands-on AI/ML project development and deployment of real-world applications.
- Infosys Springboard Certification — GAN for Cardiovascular Disease (CVD)
Hands-on machine learning model training using Generative Adversarial Networks (GANs) for medical image generation and risk analysis.
- AWS Academy Graduate — AWS Academy Cloud Foundations
Completed Amazon Web Services (AWS) training covering cloud architecture, compute, security, storage, and networking services.
- Generative AI with MongoDB & Vector Search — MongoDB
Hands-on certification in semantic vector embeddings, vector search indexes, and RAG architectures.
- AWS Educate — Machine Learning Foundations & Generative AI
Amazon Web Services (AWS) training covering machine learning pipelines, deep learning fundamentals, and GenAI concepts.
- Infosys Springboard — Robotic Process Automation (RPA) & Computer Vision 101
Automation workflows, bot configuration, and image processing fundamentals with OpenCV.

Achievements

- Winner, Project Expo – Presented and demonstrated an IoT-based Smart Irrigation System, explaining its working and impact.
- Published Patent Application (Team Lead) – Filed and published Indian Patent application for Uncertainty-Aware Multimodal Crop Health AI System (App No: 202641109078).